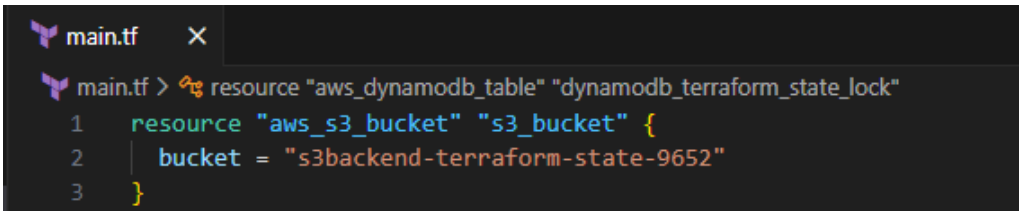
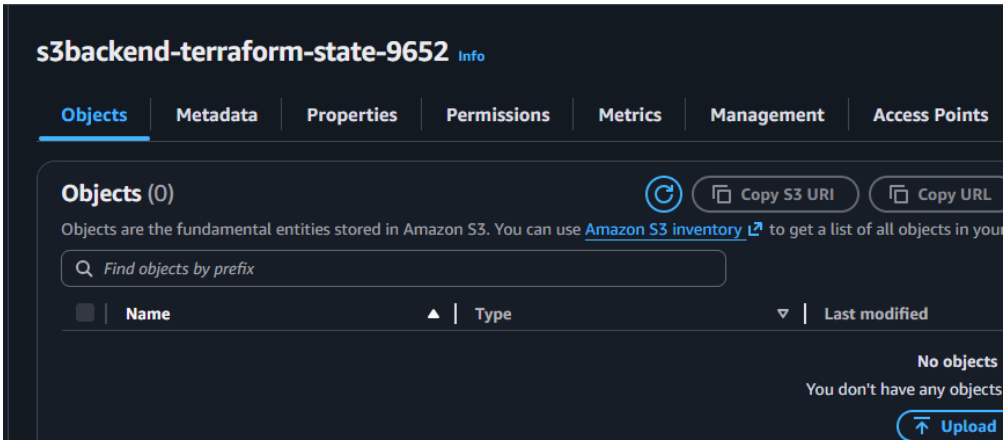
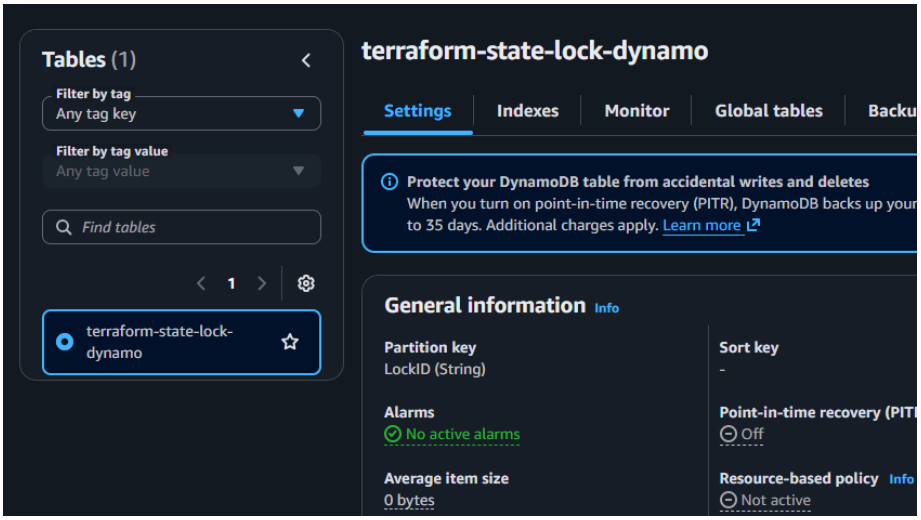
Terraform Task - 05

- Execute the script shown in the video.
- 

- I removed the ACL value since it is disabled by default, and the bucket is private by default.
- 

-

- 

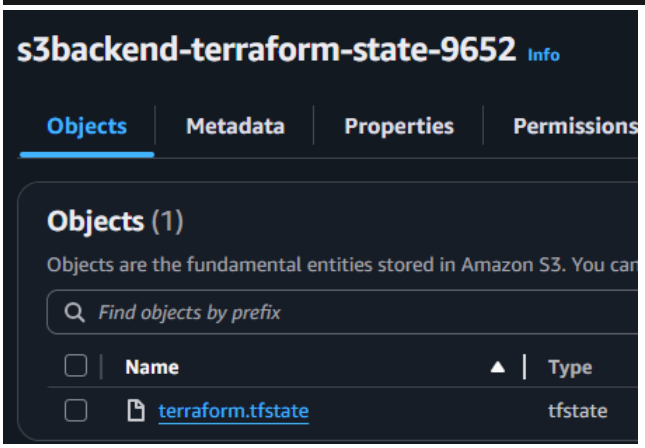
Terraform Task - 05

```
terraform {
  backend "s3" {
    bucket         = "s3backend-terraform-state-9652"
    key            = "terraform.tfstate"
    region         = "us-east-1"
    dynamodb_table = "terraform-state-lock-dynamo"
  }
}
```

-

```
PS C:\Users\Viqaas\Desktop\Testing\Terraform> terraform init
Initializing the backend...
Do you want to copy existing state to the new backend?
  Pre-existing state was found while migrating the previous "local" backend to the
  newly configured "s3" backend. No existing state was found in the newly
  configured "s3" backend. Do you want to copy this state to the new "s3"
  backend? Enter "yes" to copy and "no" to start with an empty state.

  Enter a value: yes

Releasing state lock. This may take a few moments...

Successfully configured the backend "s3"! Terraform will automatically
use this backend unless the backend configuration changes.
Initializing provider plugins...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Using previously-installed hashicorp/aws v6.34.0

Terraform has been successfully initialized!
```

-



-

```
PS C:\Users\Viqaas\Desktop\Testing\Terraform> terraform apply
Acquiring state lock. This may take a few moments...
aws_dynamodb_table.dynamodb_terraform_state_lock: Refreshing state... [id=terraform-state-lock-dynamo]
aws_s3_bucket.s3_bucket: Refreshing state... [id=s3backend-terraform-state-9652]

No changes. Your infrastructure matches the configuration.

Terraform has compared your real infrastructure against your configuration and found no differences, so no changes are needed.
Releasing state lock. This may take a few moments...

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.
PS C:\Users\Viqaas\Desktop\Testing\Terraform>
```

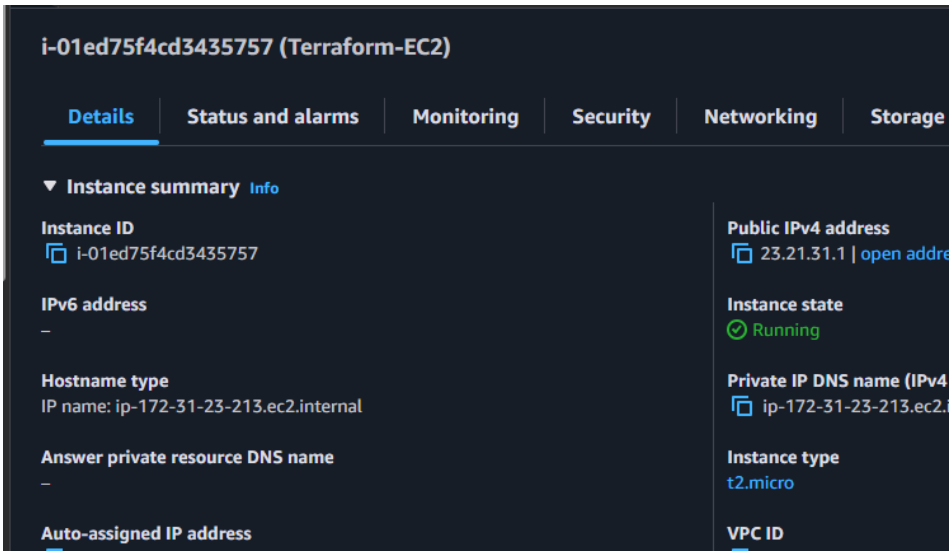
-

Terraform Task - 05

- main.tf

main.tf > resource "aws_instance" "my_ec2" > tags

```
resource "aws_instance" "my_ec2" {
  ami           = "ami-0a041b7f0a57c05b8"   # Amazon Linux (us-east-1)
  instance_type = "t2.micro"

  tags = {
    Name = "Terraform-EC2"
  }
}
```

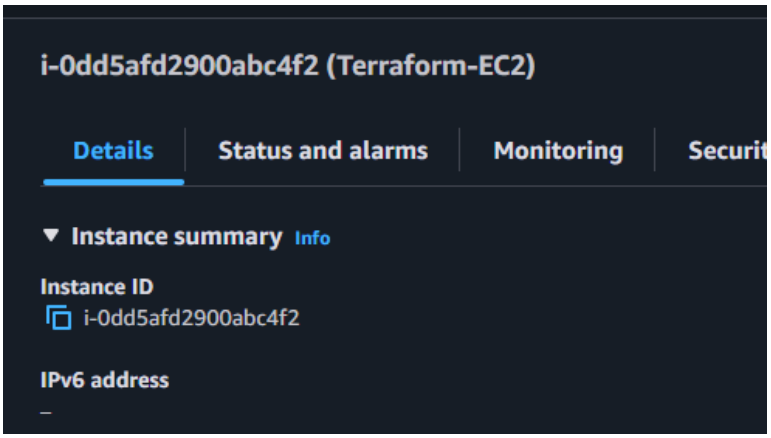
- 

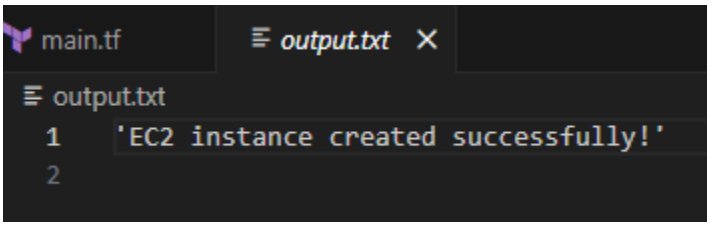
- main.tf

main.tf > resource "aws_instance" "my_ec2"

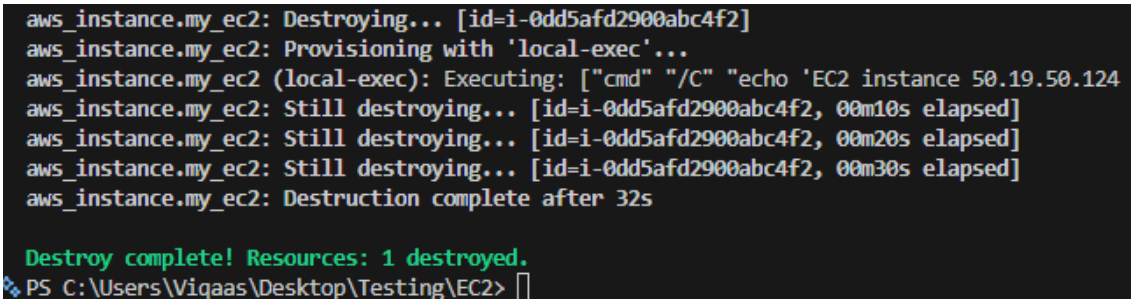
```
resource "aws_instance" "my_ec2" {
  ami           = "ami-0f4b77a23eac2378a"
  instance_type = "t2.micro"
  key_name      = "Linux-key"

  tags = {
    Name = "Terraform-EC2"
  }
  provisioner "local-exec" {
   command = "echo 'EC2 instance created successfully!' > output.txt"
  }
}
```

Terraform Task - 05

- 

- 

- During destroy, Terraform only allows access to the same resource — so we use self instead of full name.
-

- 

-

Terraform Task - 05

- main.tf > resource "aws_instance" "my_ec2"

```tf
resource "aws_instance" "my_ec2" {
  ami           = "ami-0f4b77a23eac2378a"
  instance_type = "t2.micro"
  key_name      = "Linux-key"

  tags = {
    Name = "Terraform-EC2"
  }
  provisioner "local-exec" {
    command     = "invalidcommand"
    on_failure  = continue
  }
}
```

- The command did not execute, but the instance was still created.

```
aws_instance.my_ec2: Provisioning with 'local-exec'...
aws_instance.my_ec2 (local-exec): Executing: ["cmd" "/C" "invalidcommand"]
aws_instance.my_ec2 (local-exec): 'invalidcommand' is not recognized as an internal or external command,
aws_instance.my_ec2 (local-exec): operable program or batch file.
aws_instance.my_ec2: Creation complete after 18s [id=i-0c43fe0fad3cec94b]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
PS C:\Users\Viqaas\Desktop\Testing\EC2>
```

- Terraform-EC2 i-0c43fe0fad3cec94b Running

i-0c43fe0fad3cec94b (Terraform-EC2)

Details | Status and alarms | Monitoring | Security

Instance summary Info

Instance ID
i-0c43fe0fad3cec94b

- main.tf > resource "local_file" "pet"

```tf
resource "local_file" "pet" {
  filename = "pets.txt"
  content = "Never touch dogs!"
}
```

-

```
PS C:\Users\Viqaas\Desktop\Testing\EC2> terraform taint local_file.pet
Resource instance local_file.pet has been marked as tainted.
PS C:\Users\Viqaas\Desktop\Testing\EC2> terraform apply
local_file.pet: Refreshing state... [id=21e0ad6dd47821de72a2901ddf502cb0181db46f]

Terraform used the selected providers to generate the following execution plan. Res
```

Terraform Task - 05

- It will force recreate the file with every apply command, since it is marked as tainted. After successful apply, taint is removed automatically.

```
  Only 'yes' will be accepted to approve.

  Enter a value: yes

local_file.pet: Destroying... [id=21e0ad6dd47821de72a2901ddf502cb0181db46f]
local_file.pet: Destruction complete after 0s
local_file.pet: Creating...
local_file.pet: Creation complete after 0s [id=21e0ad6dd47821de72a2901ddf502cb0181db46f]

Apply complete! Resources: 1 added, 0 changed, 1 destroyed.
PS C:\Users\Viqaas\Desktop\Testing\EC2>
```

- If tainted accidentally then we can untaint it.

```
PS C:\Users\Viqaas\Desktop\Testing\EC2> terraform taint local_file.pet
Resource instance local_file.pet has been marked as tainted.
PS C:\Users\Viqaas\Desktop\Testing\EC2> terraform untaint local_file.pet
Resource instance local_file.pet has been successfully untainted.
PS C:\Users\Viqaas\Desktop\Testing\EC2>
```

- Logs

```
PS C:\Users\Viqaas\Desktop\Testing\EC2> Set-Item -Path env:TF_LOG -Value "TRACE"
PS C:\Users\Viqaas\Desktop\Testing\EC2> Set-Item -Path env:TF_LOG_PATH -Value "terraform.log"
PS C:\Users\Viqaas\Desktop\Testing\EC2> Set-Item -Path env:TF_LOG_PATH -Value "ERROR"
PS C:\Users\Viqaas\Desktop\Testing\EC2> terraform apply -auto-approve
```

```
main.tf    ERROR
ERROR > ...
2026-03-03T15:44:41.444+0530 [INFO]  Terraform version: 1.14.6
2026-03-03T15:44:41.445+0530 [DEBUG] using github.com/hashicorp/go-tfe v1.94.
2026-03-03T15:44:41.445+0530 [DEBUG] using github.com/hashicorp/hcl/v2 v2.24.
2026-03-03T15:44:41.445+0530 [DEBUG] using github.com/hashicorp/terraform-svc
2026-03-03T15:44:41.445+0530 [DEBUG] using github.com/zclconf/go-cty v1.16.3
2026-03-03T15:44:41.445+0530 [INFO]  Go runtime version: go1.25.6
2026-03-03T15:44:41.445+0530 [INFO]  CLI args: []string{"D:\\DevOps\\terrafor
"-auto-approve"}
2026-03-03T15:44:41.445+0530 [TRACE] Stdout is a terminal of width 173
2026-03-03T15:44:41.445+0530 [TRACE] Stderr is a terminal of width 173
2026-03-03T15:44:41.445+0530 [TRACE] Stdin is a terminal
2026-03-03T15:44:41.448+0530 [DEBUG] Attempting to open CLI config file: C:\U
2026-03-03T15:44:41.448+0530 [DEBUG] File doesn't exist, but doesn't need to.
2026-03-03T15:44:41.448+0530 [DEBUG] ignoring non-existing provider search di
```

- Create one **EC2 instance** with **httpd installed** using a Terraform script.
- Set up **S3 as backend** for task 3.
- Set up **DynamoDB locking** for task 3.

Terraform Task - 05

```
terraform {
  backend "s3" {
    bucket         = "s3backend-9652"
    key            = "terraform.tfstate"
    region         = "us-east-1"
    dynamodb_table = "terraform-state-lock-9652"
  }
}

resource "aws_instance" "my_ec2" {
  ami           = "ami-0f4b77a23eac2378a"   # Amazon Linux (us-east-1)
  instance_type = "t2.micro"
  key_name      = "Linux-key"

  user_data = <<-EOF
              #!/bin/bash
              yum install httpd -y
              systemctl start httpd
              systemctl enable httpd
              echo "Hello from Terraform" > /var/www/html/index.html
              EOF

  tags = {
```

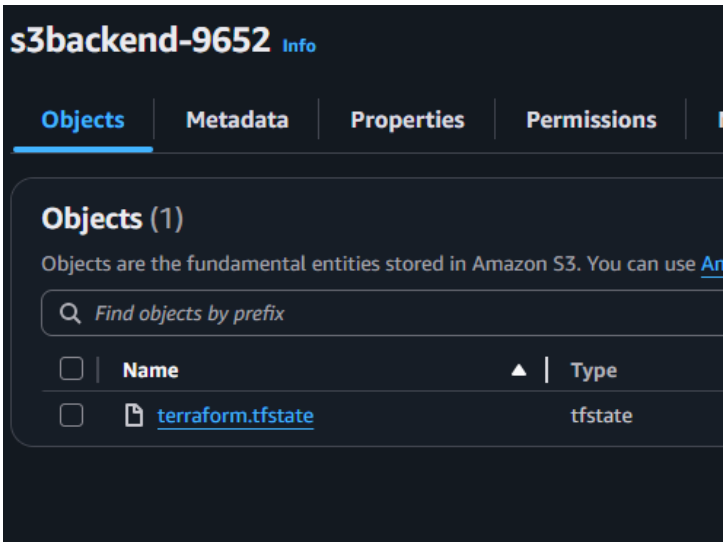
-

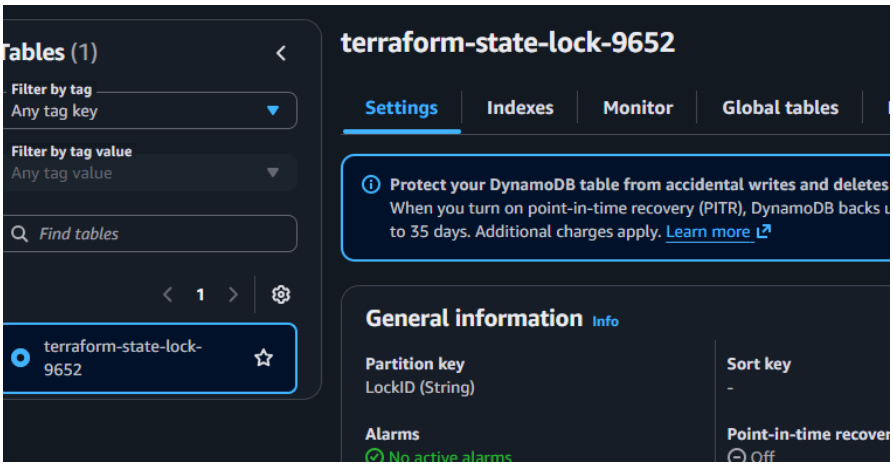
backend.tf > terraform

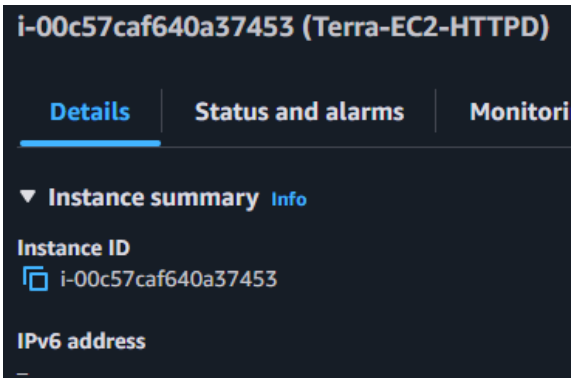
```
resource "aws_s3_bucket" "s3_bucket" {
  bucket = "s3backend-9652"
}

resource "aws_dynamodb_table" "dynamodb_terraform_state_lock" {
  name         = "terraform-state-lock-9652"
  billing_mode = "PROVISIONED"

  hash_key     = "LockID"
  read_capacity  = 20
  write_capacity = 20

  attribute {
    name = "LockID"
    type = "S"
  }
}
terraform {
  backend "s3" {
    bucket         = "s3backend-9652"
    key            = "terraform.tfstate"
    region         = "us-east-1"
    dynamodb_table = "terraform-state-lock-9652"
  }
}
```

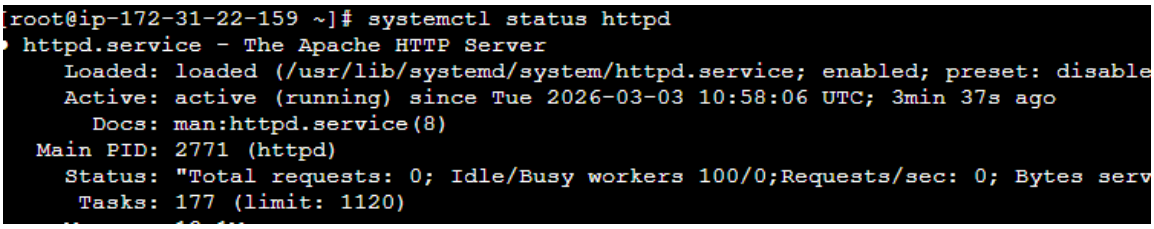
-

Terraform Task - 05

- 

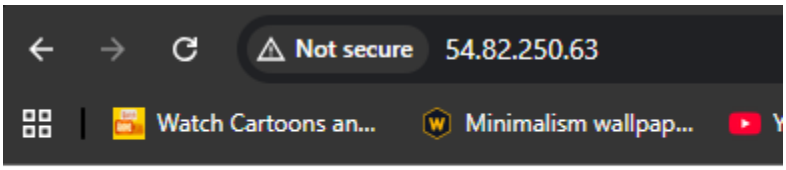
- 

- 

- 

```
root@ip-172-31-22-159 ~]# systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disable
   Active: active (running) since Tue 2026-03-03 10:58:06 UTC; 3min 37s ago
     Docs: man:httpd.service(8)
 Main PID: 2771 (httpd)
   Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes serv
    Tasks: 177 (limit: 1120)
```

Terraform Task - 05



Hello from Terraform

-